

DIFFERENTIAL COHOMOLOGY SEMINAR 12

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Today we want discuss twisted differential cohomology theory following [BG21].

1. REVIEWING TWISTED COHOMOLOGY

Let us first recall the definition of twisted cohomology theories, as discussed in a previous talk [Ber25]. Let X be a topological spaces, R be a commutative ring spectrum. We denote by Mod_R the ∞ -category of R -module spectra. The Picard ∞ -groupoid of R is defined as

$$\mathrm{Pic}_R := \mathrm{Pic}(\mathrm{Mod}_R) \subset \mathrm{Mod}_R$$

the full sub- ∞ -groupoid of invertible R -modules and equivalences.

Definition 1.1. A *twist* of R over X is a map of spaces (∞ -groupoids)

$$\alpha : X \rightarrow \mathrm{Pic}_R.$$

Definition 1.2. Given a twist $\alpha : X \rightarrow \mathrm{Pic}_R$, the α -*twisted R -cohomology* of X is defined as

$$R^\alpha(X) := \mathrm{Map}_{\mathrm{Mod}_R}(R, \alpha(X)).$$

We can now compute the twisted cohomology groups as homotopy groups of the mapping spectrum.

$$R^{k+\alpha} = \pi_k(R^\alpha(X)) = \pi_k(\mathrm{Map}_{\mathrm{Mod}_R}(R, \alpha(X))).$$

This coincides with the n -homotopy group of $\Gamma(X, \alpha)$ where $\Gamma(X, \alpha)$ is the spectrum of sections of the bundle of spectra over X associated to the twist α .

2. REVIEWING DIFFERENTIAL COHOMOLOGY

We now want to recall the definition of differential cohomology theories. Here we don't follow the more modern approach via pure and $\mathbb{R}$ -invariant sheaves of spectra, as studied in [BNV16], but rather the more classical approach [HS05]. This approach is also the one used in previous talks to study differential K -theory [Lud25].

We first review some notations and definition. Let $\mathrm{Ch}(\mathbb{R})$ be the 1-category of chain complexes of real vector spaces. We denote by $\mathcal{D}(\mathbb{R})$ the derived ∞ -category obtained from $\mathrm{Ch}(\mathbb{R})$ by inverting quasi-isomorphisms. Finally, let C be an object in an ∞ -category $\mathcal{C}$. Then $\underline{C}$ denotes the constant sheaf with value C .

Definition 2.1. Let X be a spectrum. A *differential refinement* of X is a triple $(V, c: R \wedge X \rightarrow HV)$, where

- V is a chain complex of real vector spaces,
- HV is the Eilenberg-MacLane spectrum associated to V , via the stable Dold-Kan correspondence,
- α is an equivalence of spectra.

Before we can proceed, we need to introduce further notation. Let V be a chain complex of real vector spaces. Let Ω^*V be the sheaf of differential forms with values in V .

Definition 2.2. Let $\mathcal{M}$ be a sheaf of chain complexes on the site of manifolds. The *naive truncation* $\mathcal{M}^{\geq n}$ is defined as $\mathcal{M}$ if $k > n$ and 0 otherwise.

We can now get higher categorical sheaves out of sheaves of truncated chain complexes via localizations.

Lemma 2.3. *Let $\mathcal{M}$ be a sheaf of C^∞ -modules, then post-composition with the localization functor $\mathrm{Ch}(\mathbb{R}) \rightarrow \mathcal{D}(\mathbb{R})$ preserves limits, and hence preserves the sheaf condition.*

Definition 2.4. Let (X, V, c) be a differential refinement of a spectrum X , and $n \in \mathbb{Z}$. Let $F^n(X, V, c)$, the *differential function spectrum*, be defined as the pullback

$$\begin{array}{ccc} F^n(X, V, c) & \xrightarrow{\quad\quad\quad} & H(\Omega^\bullet V^{\geq n}) \\ \downarrow & & \downarrow c \\ \underline{X} & \xrightarrow{1 \wedge X} \underline{H\mathbb{R}} \wedge \underline{X} \xrightarrow{c} \underline{HV} \longrightarrow & H(\Omega^\bullet V) \end{array}$$

Finally we can use the differential function spectrum to define differential cohomology groups.

Definition 2.5. Let X be a spectrum, (X, V, c) be a differential refinement of X , and $n \in \mathbb{Z}$. Then

$$\hat{X}^n(M) := \pi_{-n}(F^n(X, V, c)(M))$$

is the n -th *differential X -cohomology group* of the manifold M .

3. TOWARDS TWISTED DIFFERENTIAL COHOMOLOGY

We can now combine the two previous definitions to define twisted differential cohomology theories. Let R be a commutative ring spectrum, M a manifold. We now want to define sheaf theoretic analogue of a twist, generalizing our previous definitions. Here we start using non-trivial definitions and results of [BG21].

Definition 3.1. Let $\text{Mod}_{\underline{R}}(M)$ be the ∞ -category of sheaves of $\underline{R}$ -module spectra on M (i.e. sheaves valued in Mod_R). Define $\text{Pic}_{\underline{R}}^{\text{loc}}(M)$ as the full sub- ∞ -groupoid of $\text{Mod}_{\underline{R}}(M)$ spanned by the locally constant $\underline{R}$ -modules. The objects of $\text{Pic}_{\underline{R}}^{\text{loc}}(M)$ are called *topological R -twists* of M .

This name is motivated by the following observation. There is an equivalence

$$\mathcal{F}\text{un}(M^{\text{top}}, \text{Pic}_R) \rightarrow \text{Pic}_R^{\text{loc}}(M) \hookrightarrow \text{Shv}((\text{Mfd}/M)^{\text{op}}, \text{Mod}_R),$$

which sends f to the sheaf that sends $g: N \rightarrow M$ to the R -module of global sections of $f \circ g$. Here M^{top} is the underlying topological space of the manifold M . Here we also used the fact that $\text{Shv}((\text{Mfd}/M)^{\text{op}}, \text{Mod}_R)$ is equivalent to the $\text{Shv}((\text{Mfd})^{\text{op}}, \text{Mod}_R)$ objects over the locally constant sheaf $\underline{M}$. So, the topological R -twists of M are precisely the twists that come from a functor $M \rightarrow \text{Pic}_R$, which is what a twist should really mean.

We now use this notion to define twisted differential cohomology.

Remark 3.2. Recall that the localization functor $\text{Ch}(\mathbb{R})\text{Mod}_{H\mathbb{R}}$ is lax monoidal, meaning it preserves commutative algebra objects.

The previous remark implies that the composition of the stable Dold-Kan equivalence $\text{Ch}(\mathbb{R}) \rightarrow \mathcal{D}(\mathbb{R})$ with the localization $\text{Ch}(\mathbb{R})\text{Mod}_{H\mathbb{R}}$ is lax monoidal and hence preserves algebra objects. This means we get a functor

$$\mathcal{C}\text{Mon}(\text{Ch}(\mathbb{R})) \rightarrow \mathcal{C}\text{Mon}(\text{Mod}_{H\mathbb{R}})$$

Here the left hand side is the category of differential graded commutative algebras (CDGAs) over $\mathbb{R}$, while the right hand side is the category of commutative $H\mathbb{R}$ -algebras.

Definition 3.3. Let R be a commutative ring spectrum, A *differential (ring) refinement* of R is a triple (A, c, R) where

- A is a CDGA over $\mathbb{R}$,
- $c: R \wedge H\mathbb{R} \rightarrow HA$ is an equivalence of commutative $H\mathbb{R}$ -algebras.

We can in fact make this definition more categorical.

Definition 3.4. We define the ∞ -category $\widehat{\text{Ring}}$ of differential ring refinements as the pullback

$$\begin{array}{ccc} \widehat{\text{Ring}} & \xrightarrow{\quad\quad\quad} & \text{DGCA} \text{arrow}[d, "H"] \\ \downarrow & & \\ \mathcal{C}\text{Mon}(\text{Sp}) & \xrightarrow{- \wedge H\mathbb{R}} & \mathcal{C}\text{Mon}(\text{Mod}_{H\mathbb{R}}) \end{array}$$

Notice we can use the above to definition to again define a new sheaf of ring spectra via pullback.

Definition 3.5. Let (R, A, c) be a differential refinement of R . Then the *differential function ring spectrum* $\hat{R}$ is defined as the pullback

$$\begin{array}{ccc} \hat{R} & \xrightarrow{\quad} & H(\Omega^\bullet A)^{\geq n} \\ \downarrow & & \downarrow \\ \underline{R} & \xrightarrow{\quad} & \underline{R} \wedge H\mathbb{R} \xrightarrow{c} \underline{HA} \xrightarrow{\quad} & H(\Omega^\bullet A) \end{array}$$

We now want to define a notion of “differential twists” and “differential module refinements” of topological twists. For this we need to introduce some more notation and elaborate properties.

Definition 3.6. Let A be a CDGA over $\mathbb{R}$. Let $\text{Pic}_A^{wloc, fl}(M)$ be the full sub- ∞ -groupoid of $\text{Mod}_{\Omega^\bullet A}(M)$ spanned by the sheaves $\mathcal{M}$ of $\Omega^\bullet A$ -modules which are

- *weakly locally constant*, i.e. locally equivalent to a constant sheaf of $\Omega^\bullet A$ -modules, as a sheaf valued in $\mathcal{D}(\mathbb{R})$,
- *K-flat*, i.e. such that the functor $\mathcal{M} \otimes_{\Omega^\bullet A} -$ preserves quasi-isomorphisms,
- *invertible*, i.e. there exists $\mathcal{N}$ such that $\mathcal{M} \otimes_{\Omega^\bullet A} \mathcal{N} \cong \Omega^\bullet A$ as sheaves valued in $\text{Ch}(\mathbb{R})$.

We are now ready to give the main definition of twisted differential cohomology.

Definition 3.7. Let E be a topological twist in $\text{Pic}_{\underline{R}}^{loc}(M)$, a *differential module refinement* of E is a triple $(E, \mathcal{M}, c)$ where

- $\mathcal{M} \in \text{Pic}_A^{wloc, fl}(M)$,
- $c: E \wedge H\mathbb{R} \rightarrow H\mathcal{M}$ is an equivalence in $\text{Pic}_{\underline{HA}}^{loc}(M)$.

Definition 3.8. Let $(E, \mathcal{M}, c)$ be a differential module refinement of a topological twist $E \in \text{Pic}_{\underline{R}}^{loc}(M)$. Let $F(E, \mathcal{M}, c)$ be the pullback

$$\begin{array}{ccc} F(E, \mathcal{M}, c) & \xrightarrow{\quad} & H(\mathcal{M}^{\geq 0}) \\ \downarrow & & \downarrow \\ E & \xrightarrow{\quad} & \underline{H\mathbb{R}} \wedge E \xrightarrow{c} \underline{H\mathcal{M}} \end{array}$$

Denote by $\hat{R}^{(E, \mathcal{M}, c)}(M) := \pi_0(F(E, \mathcal{M}, c)(M))$ the *twisted differential R-cohomology group* of M associated to the differential refinement $(E, \mathcal{M}, c)$ of the topological twist E .

Definition 3.9. Let $\text{Tw}_{\hat{R}}(M)$, the ∞ -category of differential twists, be the ∞ -category defined as the pullback

$$\begin{array}{ccc} \text{Tw}_{\hat{R}}(M) & \xrightarrow{\quad} & \text{Pic}_{\Omega^\bullet A}^{wloc, fl}(M) \\ \downarrow & & \downarrow H \\ \text{Pic}_{\underline{R}}^{loc}(M) & \xrightarrow{- \wedge H\mathbb{R}} & \text{Pic}_{\underline{HA}}^{loc}(M) \end{array}$$

As a next step we can explore applications and examples of this abstract setup, in particular in the context of differential twisted K -theory.

4. DIFFERENTIAL TWISTED REFINEMENTS OF K -THEORY

This section is some experimental effort towards understanding twisted differential K -theory and how we get classes via bundle gerbes.

Recall that for K -theory, $GL_1 K = BS^1 \times \mathbb{Z}/2\mathbb{Z} \times BSU$, as by definition we have the pullback diagram

$$\begin{array}{ccc} GL_1 K & \xrightarrow{\quad} & \Omega^\infty KU \\ \downarrow & & \downarrow \\ \pi_0(KU)^\times & \xrightarrow{\quad} & \pi_0(KU) \end{array},$$

where $\mathbb{Z} \times BU \simeq \Omega^\infty K \simeq \mathcal{H}om_s(*, K) \simeq \mathcal{H}om_{\text{Mod } K}(K, K)$.

So for a given bundle gerbe, meaning a map $X \rightarrow K(\mathbb{Z}, 3)$, we get via post-composition with $K(\mathbb{Z}, 3) \rightarrow BGL_1 K$, a twist of K -theory. So we have bijection between homotopy classes of maps from X into $\tau_3 BGL_1 K$ (the 3-truncation of $BGL_1 K$) and homotopy classes of super bundle gerbes on X

$$[X, \tau_3 BGL_1 K] \cong \pi_0 sBldGrb(X).$$

More generally, we also have a bijection between homotopy classes of maps from X into $\tau_3 \text{Pic}_K$ (the 3-truncation of Pic_K) and homotopy classes of 2-line bundles on X

$$[X, \tau_3 \text{Pic}_K] \cong \pi_0 2LineBdl(X).$$

We now have differential refinements of $K(\mathbb{Z}, 3)$ and $BGL_1 K$ given by Deligne cohomology and the Hopkins-Singer model of differential K -theory, respectively. So we can ask whether the map $K(\mathbb{Z}, 3) \rightarrow BGL_1 K$ admits a differential refinement. Recall that this model of differential K -theory is given by the pullback

$$\begin{array}{ccc} \hat{K} & \longrightarrow & H\Omega^*(-, \mathbb{R}[u, u^{-1}]) \\ \downarrow & & \downarrow \\ \underline{K} & \longrightarrow & \underline{H\mathbb{R}[u, u^{-1}]} \end{array},$$

where $|u| = -2$.

Similarly, Deligne cohomology is given by the pullback

$$\begin{array}{ccc} \mathbb{Z}(3) & \longrightarrow & H\Omega^*(-)^{\geq 3} \\ \downarrow & & \downarrow \\ \underline{H\mathbb{Z}} & \longrightarrow & \underline{H\mathbb{R}} \end{array}.$$

Now given a commutative ring spectrum R , we can consider the corresponding differential refinements, as a DGCA A and an equivalence $c: R \wedge H\mathbb{R} \rightarrow HA$. This gives us a diagram

$$\begin{array}{ccc} \widehat{R(n)} & \longrightarrow & H\Omega^*(-, A)^{\geq n} \\ \downarrow & & \downarrow \\ \underline{R} & \longrightarrow & \underline{HA} \end{array}$$

and we have

$$\hat{R}^n(M) = \pi_{-n}(\widehat{R(n)}(M)).$$

A twist of $\hat{R} = (R, A, c)$ over X consists of

- (1) a topological twist $E\text{Sing}X \rightarrow \text{Pic}_R$,
- (2) a sheaf $\mathcal{M}$ of $\text{Pic}_{\Omega^*A}^{wloc, flat, \dots}|_X \Omega^*(-, A)|_X$ -modules on X ,
- (3) an equivalence $\tau: \underline{E} \wedge HA \rightarrow Hi\mathcal{M}$ in $\text{Pic}_{\underline{HA}}^{loc}(X)$. Here $i: \text{Ch}(\mathbb{R}) \rightarrow \mathcal{D}(\mathbb{R})$ is the localization functor. Here $\underline{E}$ is the locally constant sheaf of R -modules associated to E (i.e., the sheafification of the constant sheaf).

Let us now see how we can go from twisted Deligne cohomology to twisted differential K -theory.

Example 4.1. Let $\omega \in \Omega^3(M)$, and take $(\Omega^*(-, \mathbb{R}[u, u^{-1}]), d + u\omega)$. Now given a bundle gerbe with connection, we get a degree 3-form ω as curvature, and the bundle gerbe gives us the map $E: M \rightarrow BGL_1 K$. From the perspective of the definition

- (1) $\hat{R} = \hat{K} = (K, \mathbb{R}[u, u^{-1}], \text{chern character})$
- (2) E is obtained from the bundle gerbe, as explained above,
- (3) $\mathcal{M}$ is the sheaf of $\Omega^*(-, \mathbb{R}[u, u^{-1}])$ -modules given by $(\Omega^*(-, \mathbb{R}[u, u^{-1}]), d + u\omega)$, and τ is the equivalence given by the Chern character, which is an equivalence of sheaves of $H\mathbb{R}$ -modules, as the Chern character is an equivalence after tensoring with $\mathbb{R}$.

Note here we have the computation on $(\Omega^*A, d + \lambda)$

$$(d + u\omega)(ax) = da \cdot x \pm a \cdot dx + u\omega ax = da \cdot x \pm a(d + u\omega)x$$

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